Department of Computer Science & Engineering
School of Engineering & Technology
Guru Ghasidas Vishwavidyalaya (Central University)
(B.tech 2nd year examination, 4th SEM)
CT-2 Examinations – April, 2024

Subject: **Introduction to Information Science** Branch: **CSE** Duration: 1 Hour
Date: **30/04/2024** Paper code: **CSUDT01** Max Marks: 15

Note:-**Attempt all questions. All Questions Carries Equal Marks**

a) Calculate the channel capacity using the Shannon-Hartley theorem when the communication channel is with the following parameters: channel bandwidth=4khz, signal power=2mW, and Noise power=0.5mW.**[CO-2, BTL-1, Marks-5]**

b) Define with one example:- **[CO-2, BTL-1, Marks-5]**
   a) Parity check
   b) Single-bit error
   c) Multiple-bit error
   d) Burst error
   e) Block codes

c) Write down any five differences between the Lossless and Lossy compression schemes.**[CO-3, BTL-1, Marks-5]**

SET-A

SECTION-B

Note: Attempt any one question from each unit. [8x5=40]

UNIT-I

1. Describe the following channels with figure: [CO-1, BLT-2, Marks-8]
   (a) Lossless channel
   (b) Deterministic channel
   (c) Noiseless channel
   (d) Binary symmetric channel
2. Apply Shanon-fano coding for the following message [CO-1, BLT-2, Marks-8]
   $[M]=[M_1,M_2,M_3,M_4,M_5,M_6],[P]=[0.30,0.25,0.15,0.12,0.08,0.10]$
   Calculate the value of
   (a) Entropy
   (b) Average Length
   (c) Efficiency
   (d) Redundancy

UNIT-II

3. Define CRC? Explain it with one numerical example. [CO-2, BLT-2, Marks-8]
4. Calculate the channel capacity using the Shanon-Hartley theorem when the communication channel is with the following parameters: channel bandwidth=4 kHz, signal power=2mW, and Noise Power =0.5 mW. [CO-2, BLT-2, Marks-8]

UNIT-III

5. Explain Huffman coding with one numerical example. [CO-3, BLT-2, Marks-8]
6. Write down any four differences between Lossless and Lossy Compression. [CO-3, BLT-2, Marks-8]

UNIT-IV

7. Write down any eight differences between video compression and audio compression. [CO-4, BLT-2, Marks-8]
8. Describe the need for full-motion video compression. [CO-4, BLT-2, Marks-8]

UNIT-V

9. Describe the RSA algorithm with one numerical example. [CO-5, BLT-2, Marks-8]
10. Write a short note on:- [CO-5, BLT-2, Marks-8]
    (a) Block cipher code
    (b) Stream cipher
    (c) Asymmetric key cryptography
    (d) Digital Signature

SET-A

B. Tech (Second Year (Fourth Sem) Examination 2023-24
Branch: Common for Chemical Engg. & Information Technology
Sub: Introduction to Information Science, Paper Code: CSUDT01
Time allowed: Three Hours
Maximum Marks: 60

Note: Attempt all sections as directed. Section A is compulsory.

SECTION-A

Note: All questions are compulsory.

I. **Multiple Choice Question:** **[5x2=10]**

1. The unit of average mutual information is [CO-1, BLT-1, Marks-2]
   (a) Bytes per Symbol (b) Bytes
   (c) Bits per symbol (d) Bits

2. Which is more efficient? [CO-2, BLT-1, Marks-2]
   (a) Parity Check (b) CRC
   (c) Parity & CRC (d) None of these

3. Which of the following is termed as "reducing image data"? [CO-3, BLT-2, Marks-2]
   (a) Luminance (b) Compression
   (c) Hue (d) Saturation

4. What is the full form of JPEG? [CO-4, BLT-2, Marks-2]
   (a) Joint Pressure Expert Group (b) Joint Photographic Expert Group
   (c) Joint Picture Expert Group (d) Join Pic Expert Group

5. What is Data Encryption Standards (DES)? [CO-5, BLT-2, Marks-2 Marks]
   (a) Block Cipher (b) Stream Cipher
   (c) Bit Cipher (d) Byte Cipher

II. **Define the following terms:** **[5x2=10]**

1. Entropy [CO-1, BLT-1, Marks-2 Marks]
2. Convolution Codes [CO-2, BLT-1, Marks-2 Marks]
3. Lossy Compression [CO-3, BLT-1, Marks-2 Marks]
4. Audio Compression [CO-4, BLT-1, Marks-2 Marks]
5. Cryptanalysis [CO-5, BLT-1, Marks-2 Marks]